

LiNGAM-SF: Causal Structural Learning Method With Linear Non-Gaussian Acyclic Models for Streaming Features

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Abstract—The causal structure learning for streaming features (CSLSFs) faces the following challenges: 1) the precision of learned causal structures is limited due to the score-based learning method and 2) they fail to detect the latent confounders. To address the challenges, this article proposes a novel causal structure learning method with linear non-Gaussian acyclic models for streaming features (LiNGAM-SFs), which utilizes the causal identifiability of the data. It is the first time to utilize LiNGAMs for online causal structure learning. First, we utilize the classical SF algorithm to learn the causal skeleton. This article provides the property of this skeleton, proving that two adjacent variables on an edge is one of three possible structures. Second, we give two propositions and identify the causal directions in the presence of latent variables (ICDPLV) subalgorithm to distinguish among the three structures and precisely identify the causal directions. In addition, the subalgorithm can output a candidate set of latent confounders from a local perspective. Finally, the detecting latent confounder (DLC) subalgorithm detects the latent confounders in the candidate set with a global perspective. The precision of the proposed method is increased at least by 11% on average than those of the state-of-the-art method. Furthermore, the experiments verify that the LiNGAM-SF method is able to detect the latent confounders.

Index Terms—Causal identifiability, causal structure, latent confounders, linear non-Gaussian acyclic model (LiNGAM), streaming features (SFs).

NOMENCLATURE

Notation	Description
G_t	Causal structure at the time t .
G_t'	Causal skeleton at the time t .
V_t	Observed variable set at the time t , where $v_i \in V_t$ denotes a observed feature variable.
E	Directed edge set, where $\langle v_i, v_j \rangle \in E$.

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U	Undirected edge set, where $(v_i, v_j) \in U$ needs to have the directions identified in the subsequent steps.
X_o	Observed data composed of the data x_i on the variable $v_i \in V_t$, distinguishing from the dataset X of all variables under the full observation.
$r_i^{(j)}$	Residual of regressing x_i on x_j .
A	Adjacency matrix representing the causal strength among the variables, whose element $a_{i,j}$ refers to the intensity of the direct causal impact of v_j on v_i .
$\widetilde{a_{i,j}}$	Total product of the path coefficients from v_j to v_i .
N	Noise term, where $n_i \in N$ represents the noise on v_i .
L^t	Latent confounders detected at the time t .
$v_i \perp\!\!\!\perp v_j$	Indicating that v_i and v_j are independent.
$\text{sig}_i, \text{sig}_2, r$	Hyperparameter.

I. INTRODUCTION

AS THE attention to causality increases, people start to learn the causal structure from observational data. Currently, most research on causal structure learning focuses on static data [1], [2], [3]. However, in many real-world applications [4], [5], [6], there are dynamic data where the number of samples stays constant, while the features keep growing, called streaming features (SFs) [7]. Although the causal structure learning method for static data can provide results based on all features that have arrived, it requires relearning each time the features increase with lower efficiency. Hence, people carry out research on the causal structure learning for SFs (CSLSFs). The research aims to only update the historical causal structure when new features arrive, instead of relearning.

In recent years, people have begun to pay attention to CSLSF [8], [9], [10], [11], [12]. These studies employ the scoring-based learning method to select the causal structure as the desired outcome. Fig. 1(a) indicates the unreliability of scoring. For instance, on synthetic data, we utilize the widely used MDL scoring function whose preference lies in achieving lower scores. The score of the real causal structure is close to that of Structure 2 (edge reversal) and higher than that of Structure 6 (edge missing). Therefore, blindly applying

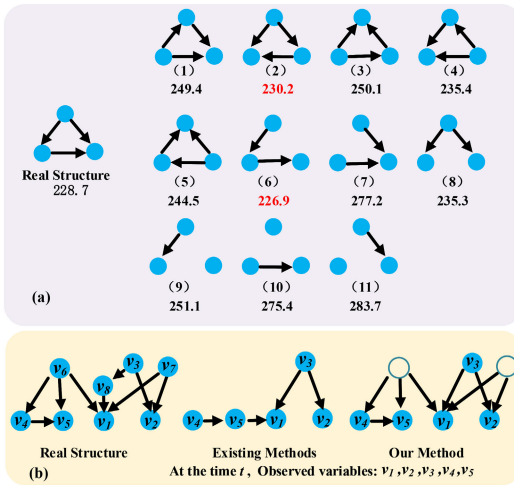


Fig. 1. Limitations of the existing CSLSF methods. (a) Scores of the structures formed by three variables on synthetic data. (b) Ignoring the existence of latent confounders.

scoring does not yield a precise causal structure. In addition, the existing methods assume that all feature variables are observed at any time, neglecting the presence of latent confounders. A latent confounder is an unobserved variable (latent variable) influencing multiple observed variables. In Fig. 1(b), the feature variables $\{v_1, v_2, v_3, v_4, v_5\}$ are observed at the time t , while the unobserved variables v_6 and v_7 are latent confounders. The existing CSLSF methods ignore the latent confounders $\{v_6, v_7\}$ and fail to detect them. They focus solely on the causal relationships among observed variables. In summary, there are two pressing issues in CSLSF methods that need resolution: 1) how to learn a precise causal structure and 2) how to detect latent confounders.

This article addresses the issues in existing CSLSF methods, focusing on linear non-Gaussian data. Many real-world datasets, such as those concerning stocks, rainfall, and functional magnetic resonance imaging (fMRI) recordings, conform to the assumption of linear non-Gaussianity [13]. Even in the cases of nonlinearity, the datasets may be approximated or modified to fit a linear relationship [14]. Besides being widely prevalent, linear non-Gaussian data also demonstrate causal identifiability [15]. Causal identifiability implies that causal relationships can be uniquely identified from observed data. Owing to the property, the linear non-Gaussian acyclic model (LiNGAM) and its various variants have been successful in causal discovery. Therefore, this article utilizes the causal identifiability of the data and proposes a method called LiNGAM-SF to solve the problems of existing CSLSF methods. As far as we know, it is the first time to dynamically learn causal structures based on LiNGAM.

This article first proves that the skeleton learned by the classical algorithm exhibits Property 1, which states that each edge's two feature variables can assume three possible structures: 1) $v_i \rightarrow v_j$ or $v_j \rightarrow v_i$; 2) $v_i \leftarrow l_k \rightarrow v_j$, where l_k is a latent confounder; and 3) $v_i \leftarrow l_k \rightarrow v_j$, and $v_i \rightarrow v_j$ or $v_j \rightarrow v_i$, where l_k is a latent confounder. Next, the proposed LiNGAM-SF provides two propositions and identifies the causal directions in the presence of latent variables (ICDPLV) subalgorithm based on Property 1 to precisely learn the causal structure. The precision achieved by the proposed

method is increased at least by 11% on average compared to those of the state-of-the-art method (see Section IV-C). The ICDPLV subalgorithm also provides a candidate set of latent confounders from a local perspective. To further find the latent confounders, the method designs the detecting latent confounder (DLC) subalgorithm to filter and detect the latent confounders from the candidate set with the global perspective. The proposed method can detect latent confounders in all designed experimental cases (see Section IV-D).

The contributions of this article are given as follows.

- 1) This article provides Property 1, which proves that two adjacent feature variables on the edges of the skeleton graph only be one of three structures.
- 2) Based on Property 1, this article proposes two propositions to precisely learn the structure by distinguishing the structure formed by the adjacent variables. Particularly, the scope of Proposition 1 expands beyond the limitation that latent variables must only include latent confounders.
- 3) This article gives Property 2, describing the attributes of p feature variables with the common confounder. Property 2 helps the DLC subalgorithm to identify the common latent confounders without relying on complex high-level statistical information.
- 4) As far as we know, it is the first exploration in the CSLSF method to detect the latent confounders. The source codes of the methods are available at <https://github.com/yuhongcqupt/LiNGAM-SF>.

The rest of this article is organized as follows. Section II describes the existing causal structure learning methods. Section III describes the proposed LiNGAM-SF method in detail. Section IV contains experiments and analysis. Section V provides conclusions and further work.

II. RELATED WORK

A. Causal Structure Learning Methods With Latent Variables

With the rise of causal learning, people have started to solve various problems in practice. For instance, Liang et al. [16] made the first attempt in attribute heterogeneous networks. Fang et al. [17] broke through the difficulty of dense and high-dimensional graph structures using low-rank technology. However, besides these, latent variables are a long-standing problem.

In the presence of latent variables, the constrained methods are still considered a favorable option because of their efficiency. Spirtes et al. [18] proposed the fast causal inference (FCI) method based on d-separation. Subsequently, Richardson [19] proposed the optimization method called RFCI based on the FCI to learn the causal structure faster. Since the above methods rely on conditional independence (CI) tests, they are highly time-consuming to deal with high-dimensional data. Therefore, several studies have commenced decreasing the quantity of CI tests. The MBCS* algorithm [20] employs the technique of Markov blankets, while the L-MARVEL method [21] removes the specific variables in the iterative process of updating causal structure. However, these constraint-based methods are unable to distinguish Markov equivalence classes, leading to the existence of undirected edges within the obtained causal structure.

The score-based methods can avoid the above issue. Triantafyllou and Tsamardinos [22] transformed the learning causal structure into a combinatorial optimization problem and obtained the causal structure by searching with the score. To further improve the efficiency, Tsirlis et al. [23] first utilized CI tests to limit the search space and then obtained the final structure through the greedy search. Bellot and van der Schaar [24] design a novel scoring method for learning the causal structure with consideration of latent confounders.

In addition to the score-based methods, the functional causal models are also capable of distinguishing Markov equivalence classes. However, these methods need to fulfill some assumptions. The assumption of Gaussian linear acyclicity is crucial as it signifies the causal identifiability within the data. Any models meeting this assumption can be classified within the LiNGAM framework. LiNGAM [15] and the variant DirectLiNGAM [13] are able to identify the causal order without latent variables. In the presence of latent variables, the LvLiNGAM [14], the ParceliNGAM [25], the MLCLiNGAM [26], and Salehkaleybar's method [27] are able to not only identify the causal order of variables but also detect latent confounders.

B. CSLSF Methods

Although all the above methods have their own advantages, none of them can cope with SFs well. The reason is that most of the learning methods for the causal structure with latent variables mainly focus on static scenarios and are one-time learning methods. In the real scenarios of SFs, new features are continuously added to the observations over time. Therefore, the above methods will be relearned once new features arrive. Not exploiting the information of the historical causal graphs leads to repeated learning with considerable time cost. Hence, the CSLSF method aims at reducing the learning time in SFs by utilizing historical causal graphs.

Yu et al. [8] conducted research on SFs and first proposed causal structure learning methods CDFSf and S-CDFSf. However, they cannot identify the causal order among variables. For this reason, Guo and Yang [9] proposed an optimized method called CSBS, but too many CI tests limit the performance of the method. To learn the causal structure faster, Yang et al. [10] proposed a new method CSSU based on symmetric uncertainty, but the method is not good at dealing with nonlinear causality. Subsequently, Yang et al. [11] investigated the data with nonlinear causality and presented the CANSF method based on the additive noise model. Unlike the previous two methods, the PRCDsf method [12] is able to handle linear and nonlinear causality. However, it still uses scoring to identify the causal direction and learn imprecise causal structure. Notably, besides studying dynamic scenarios of SFs, You et al. [28] initially study causal structure learning with incremental changes in both instances and features, introducing the C²SL_{IFS} method.

In order to illustrate the advantages and disadvantages of each method clearly, the characteristics of CSLSF methods are summarized in Table I.

From Table I, it can be found that most existing CSLSF methods learn the structure by scoring. However, the score-based methods face challenges in learning a precise

TABLE I
COMPARISON OF CSLSF METHODS

Methods	Character 1		Character 2		Character 3	
	Scoring	Others	Yes	No	Linear	Nonlinear
CDFSf and S-CDFSf [8]	-	-		✓	✓	✓
CSBS [9]	✓			✓	✓	✓
CSSU [10]	✓			✓	✓	
CANSF [11]	✓			✓		✓
PRCDsf [12]	✓			✓		✓
The proposed method		✓	✓		✓	

Character 1 refers to "The way to learn the causal structure", Character 2 indicates "Whether the latent confounders can be detected", and Character 3 refers to "The causal relationship between the variables".

structure, as shown in Fig. 1(a). In addition, the existing methods ignore the latent variables, and they are unable to detect the latent confounders, as illustrated in Fig. 1(b). Therefore, a new CSLSF method is proposed for linear non-Gaussian data.

III. METHODOLOGY

A. Problem Definition

Let $G = (V, E)$ denote a causal structure, where V refers to the set of nodes and indicates the set of features and E is the set of directed edges on the graph. The edge $\langle v_i, v_j \rangle \in E$ denotes that variable $v_i \in V$ has a causal effect on variable $v_j \in V$.

The LiNGAM-SF method proposed in this article satisfies the following basic assumptions.

- 1) The causal structure G is a directed acyclic graph.
- 2) Causal faithfulness assumption.
- 3) Causal Markov assumption.
- 4) The causal relationship among the variables is linear, and their data obey the non-Gaussian distribution.
- 5) The noise terms follow the non-Gaussian distribution and are independent of each other.

It is assumed that the data set $X \in \mathbf{R}^{n \times d}$ is obtained with the full observations, where n is the number of samples and d denotes the dimension of the feature space under full observation. The data on the node $v_i \in V$ are denoted as $x_i = X_{:,i}$. Since the data generation mechanism is assumed to be linearly non-Gaussian, the structural equation can be given

$$X^T = AX^T + N \quad (1)$$

where A is the adjacency matrix of G and $N \in \mathbf{R}^{d \times n}$ is the noise term of the corresponding variable. The element $a_{i,j}$ refers to the strength of the direct causal effect of $v_j \in V$ on $v_i \in V$.

For SFs, the observed features are not always complete. Assuming that at the time t , the data $X_o \in \mathbf{R}^{n \times d_o}$ from the set of variables $V_t \subset V$ are observed, where $d_o = |V_t|$. As time progresses, new variables are introduced through streams, contributing to the expansion of the observed data's dimension d_o . The data for the unobserved variables $V_l \subset V$ are expressed as $X_l \in \mathbf{R}^{n \times d_l}$, where $d_l = |V - V_t|$. According to the method in the literature [27], (1) can be transformed as follows:

$$\begin{bmatrix} X_o^T \\ X_l^T \end{bmatrix} = \begin{bmatrix} A_{oo} & A_{ol} \\ A_{lo} & A_{ll} \end{bmatrix} \begin{bmatrix} X_o^T \\ X_l^T \end{bmatrix} + \begin{bmatrix} N_o \\ N_l \end{bmatrix} \quad (2)$$

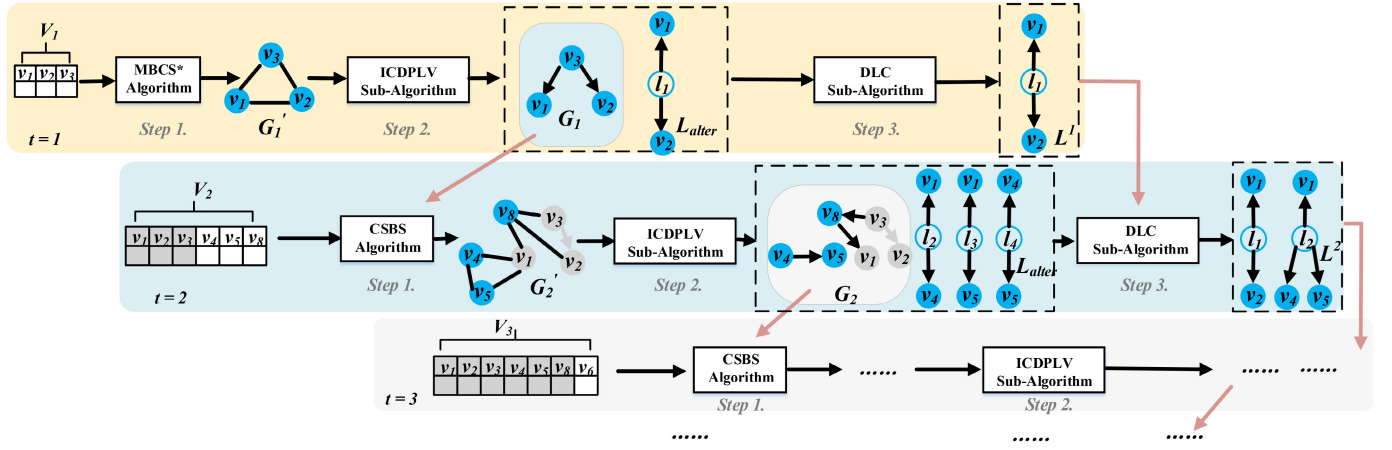


Fig. 2. Framework of the LiNGAM-SF, which is summarized in three steps. *Step 1*: learning the causal skeleton G_t' (lines 2–6 of Algorithm 1). *Step 2*: transforming the causal skeleton G_t' into a directed graph G_t and providing a candidate set of latent confounders (line 7 of Algorithm 1). *Step 3*: detecting the latent confounders L^t (line 8 of Algorithm 1).

where N_o and N_l denote the noise terms on the observed variables V_t and the unobserved variables $V - V_t$, respectively. The matrix $[A_{oo}, A_{ol}; A_{lo}, A_{ll}]$ in (2) corresponds to A in (1).

Based on (2), X_o^T can be written as follows:

$$X_o^T = B_o N_o + B_l N_l = B N \quad (3)$$

where $B_o = (I - A_{oo} - A_{ol}(I - A_{ll})^{-1}A_{lo})^{-1}$ and $B_l = (I - A_{oo} - A_{ol}(I - A_{ll})^{-1}A_{lo})^{-1}$. Akbari et al. [21] explain that the element of matrix B in (3) indicates whether there is a causal path between the corresponding variables.

At the time t , the current set of feature variables is V_t , and the causal structure is denoted as G_t . The LiNGAM-SF method is to learn the causal structure G_t from the observable data X_o .

To ensure clarity, the important notations employed in this article are summarized in Nomenclature.

B. Framework of LiNGAM-SF Method

Compared with the existing CSLSF methods, the LiNGAM-SF method can both give a precise causal structure and detect latent confounders. To achieve this goal, this article designs the overall framework of the LiNGAM-SF method. Fig. 2 shows the LiNGAM-SF framework presented with the real structure of Fig. 1(b) as an example. In the frame diagram, the undirected graph or a locally directed graph at the time t can be expressed as G_t' . The graph $G_t' = (V, E, U)$ is also known as the causal skeleton, where U represents the set of undirected edges. The undirected edge is denoted as $(v_i, v_j) \in U$, indicating that the causal direction between v_i and v_j needs to be further identified. The pseudocode of the whole process is given in Algorithm 1.

C. Learning of Causal Skeleton

The causal skeleton is an undirected graph constructed with feature variables. Unlike the maximal ancestral graph, the partial ancestral graph, and the directed acyclic graph, it does not contain directional information. Our method employs the MBCS* algorithm [20] and the CSBS algorithm [9] to learn

Algorithm 1 LiNGAM-SF Algorithm

Input: The causal structure G_{t-1} ; The observed data X_o ; The feature sets V_{t-1} and V_t ; The sampling rate r ; Two significant coefficients sig_1, sig_2

Output: The causal structure G_t ; The set of latent confounders L^t

- 1: Initial $L^t \leftarrow \emptyset$ and $G_t \leftarrow \emptyset$;
- 2: **if** V_{t-1} is Empty **then**
- 3: $G_t' \leftarrow \text{MBCS}^*(V_t, X_o, sig_1)$
- 4: **else**
- 5: $G_t' \leftarrow \text{CSBS}(V_{t-1}, V_t, X_o, G_{t-1}, sig_1)$
- 6: **end if** $\triangleright$ The code in lines 2-6 is step 1
- 7: $G_t, L_{alter} \leftarrow \text{ICDPLV}(G_t', X_o, r, sig_2)$ $\triangleright$ step 2
- 8: $L^t \leftarrow \text{DLC}(G_t, L_{alter}, X_o)$ $\triangleright$ step 3
- 9: **return** G_t and L^t

the causal skeleton G_t' without any steps to identify the direction.

The MBCS* algorithm, a static method, is used for learning the initial skeleton G_1' due to its efficiency. As new variables become observable, the CSBS algorithm dynamically updates the skeleton G_t' , $t \in \{2, 3, 4, \dots\}$. Supposed that the neighbor set of variable v_i on the skeleton is denoted as $N(v_i)$, the update requires that v_i and its neighbor $v_j \in N(v_i)$ satisfy the condition: $v_i \not\perp\!\!\!\perp v_j | \emptyset$ and $\neg \exists s \subseteq N(v_i)$ s.t. $(v_i \perp\!\!\!\perp v_j | s)$ [9]. Notably, the directed edges of G_t' are treated as undirected in each update. As shown in Fig. 2, the transformation from G_1 to G_2' is a process during which the CSBS algorithm updates the skeleton. The CSBS treats the directed edge $v_3 \rightarrow v_1$ in the G_1 as an undirected edge. As the variable v_8 is observable, v_1 and v_3 become independent given v_8 . Therefore, the directed edge $v_3 \rightarrow v_1$ is updated to the undirected path $v_3 - v_8 - v_1$. Both algorithms rely on CI tests. In our paper, we adopt the HSCI [29] with the significance parameter sig_1 .

Definition 1 (Observed Direct Path): Assume that there is a directed path $v_i \rightarrow z_1 \rightarrow z_2 \rightarrow \dots \rightarrow z_k \rightarrow v_j$ in the graph. If all variables $\{z_1, z_2, \dots, z_k\}$ on the path are latent variables, the path is called the observed direct path.

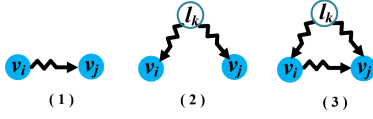


Fig. 3. Three possible structures.

We present the following three structures according to Definition 1.

- 1) *Structure 1*: There is an observed direct path between v_i and v_j , with neither variable affected by the latent confounders [see Fig. 3(a)]. It can be represented as $v_i \rightarrow v_j$ or $v_j \rightarrow v_i$.
- 2) *Structure 2*: Variables v_i and v_j are associated only due to confounding factor l_k without any observed direct path between them [see Fig. 3(b)]. It can be denoted as $v_i \leftarrow l_k \rightarrow v_j$.
- 3) *Structure 3*: There is an observed direct path between v_i and v_j ; both variables are affected by the latent confounder l_k [see Fig. 3(c)]. It can be represented as $v_i \leftarrow l_k \rightarrow v_j$ and $v_i \rightarrow v_j$ or $v_j \rightarrow v_i$.

Property 1: Two adjacent feature variables on the edge (v_i, v_j) of the skeleton graph G_t' only be one of the three structures mentioned above.

Proof: When $t = 1$, the MBCS* is used to learn the initial skeleton. The literature [20] has demonstrated that the edge $(v_i, v_j) \in U$ in the G_0' satisfies the form $v_i \leftrightarrow v_j$, indicating that no set can m-separate v_i from v_j .

When $t = 2, 3 \dots$, it can be proved by the inverse method that (v_i, v_j) in the G_t' learned by the CSBS also satisfies the form $v_i \leftrightarrow v_j$. Assuming that $(v_i, v_j) \in U$ does not follow the form $v_i \leftrightarrow v_j$, there are two cases to be discussed separately.

- 1) v_i is not directly connected to v_j , and there is a undirected path between v_i and v_j that include at least one observed collider. According to the d-separation, it is known that the collider will block the paths of the feature variables v_i and v_j . Therefore, v_i and v_j are independent. The conclusion contradicts the condition $v_i \not\perp\!\!\!\perp v_j | \emptyset$.
- 2) v_i is not directly connected to v_j , and there are no colliders among the variables on the path between v_i and v_j . When there is no collider on the path, the relation between v_i and v_j is either $v_i \leftarrow z_1 \leftarrow \dots \leftarrow z_{t-1} \leftarrow z_t \rightarrow z_{t+1} \rightarrow \dots \rightarrow z_m \rightarrow v_j$, where z_t is a confounder or a chain structure such as $v_i \rightarrow z_1 \rightarrow \dots \rightarrow z_m \rightarrow v_j$. For both structures above, the paths of the feature variables v_i and v_j are blocked once under $z_1 \in N(v_i)$ or $z_m \in N(v_j)$. The situation is in contradiction with the condition $v_i \not\perp\!\!\!\perp v_j | \emptyset$ and $\neg \exists s \subseteq N(v_i)$ s.t. $(v_i \perp\!\!\!\perp v_j | s)$.

In summary, the edge $(v_i, v_j) \in U$ on G_t' satisfies the form $v_i \leftrightarrow v_j$. The form reveals that variables v_i and v_j are not independent. Moreover, their dependence remains unchanged under other observed variables. Hence, v_i and v_j are associated through a path of latent variables, none of which are colliders. When there is no confounder on the path, the relationship between v_i and v_j belongs to structure 1. When there is exactly one latent variable acting as a confounder on the path, the relationship belongs to structure 2. When there are at least two confounders among the latent variables on the path, a collider

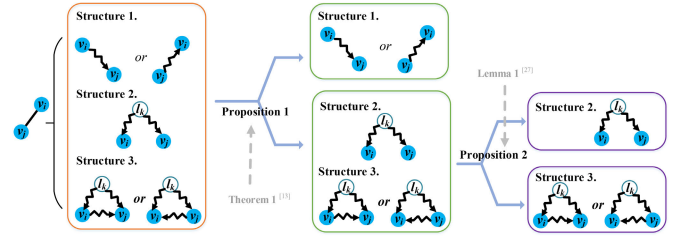


Fig. 4. Procedure of the ICDPLV subalgorithm.

must exist, and the relationship does not satisfy the form $v_i \leftrightarrow v_j$. Otherwise, the relationship between v_i and v_j can be a combination of structures 1 and 2. Therefore, our property is confirmed. $\square$

D. ICDPLV Algorithm

The graph $G_1', G_2', \dots, G_t'$ represents the learned skeleton structure. To more precisely confirm the directed structures of the graph, we propose the ICDPLV subalgorithm based on the causal identifiability of the data. The subalgorithm distinguishes among the three previously described structures to provide the precise causal structure. It also gives a candidate set of latent confounders L_{alter} . We denote a latent confounder using the variables it affects. For instance, in Structure 2 and Structure 3, the latent confounder l_k is indicated by $[v_i, v_j]$. The whole procedure of the subalgorithm is shown in Fig. 4.

1) *Proposition 1*: Proposition 1 is prepared to exclude the possibilities of Structure 2 and Structure 3 affected by latent confounders and to identify the causal direction of two variables in Structure 1. Although existing research has provided lemmas and propositions for identifying causal directions and DLCs, their assumptions require that the latent variables must only include latent confounders. They are unable to handle the situation where latent mediators v_8 are present in Fig. 1(b). Therefore, we extend the existing lemmas and propositions and present Proposition 1.

Theorem 1 (D-S Theorem [13]): Define two random variables y_1 and y_2 as linear combinations of independent random variables s_i for $i = 1, \dots, q$

$$y_1 = \sum_{i=1}^q b_i s_i, \quad y_2 = \sum_{i=1}^q c_i s_i. \quad (4)$$

If there is a non-Gaussian s_i with $b_i c_i \neq 0$, y_1 and y_2 are dependent.

Based on Theorem 1, this article gives the following proposition to identify Structure 1.

Proposition 1: Assuming that the existing assumptions are fulfilled, for the edge (v_i, v_j) in the graph G_t' , the residual is denoted as $\mathbf{r}_i^{(j)} = \mathbf{x}_i - (\text{cov}(\mathbf{x}_i, \mathbf{x}_j) / \text{var}(\mathbf{x}_j)) \times \mathbf{x}_j$. If the residual is independent of the data $\mathbf{x}_j$ of the variable v_j , then $v_j \rightarrow v_i$, and the two variables are not affected by the latent confounders.

To clarify the proof of Proposition 1, we provide the following definition.

Definition 2 (Branch Variable Set): Assuming that all variables are fully observed, the directed path $v_i \rightarrow z_1 \rightarrow \dots \rightarrow z_t \rightarrow \dots \rightarrow z_k \rightarrow v_j$ in the graph G is denoted as $P(v_i, v_j)$. For any variable z_t on the path $P(v_i, v_j)$, the

branch variable set of $H(P(v_i, v_j), z_t)$ is defined as $Pa(z_t) \setminus \{v_i, z_1, \dots, z_k, v_j\}$, where $Pa(z_t)$ represents the parent variables of z_t . The overall branch variable set for the path $P(v_i, v_j)$ is given by $H(P(v_i, v_j)) = \bigcup_{t=1}^k H(P(v_i, v_j), z_t)$.

Proof: We can assume that either $v_i \rightarrow v_j$ or both variables are influenced by common latent confounders L . For any $x_i \in \mathbf{x}_i$ and $x_j \in \mathbf{x}_j$, all paths from v_i to v_j can be expressed with a structural equation as $\widetilde{a}_{j,i}x_i + \sum_{h \in H(v_i, v_j)} \widetilde{a}_{j,h}x_h + \widetilde{n}_1$, where $\widetilde{a}_{j,i}$ and $\widetilde{a}_{j,h}$ denote the total product of path coefficients along $P(v_i, v_j)$ and from the branch variables set $H(P(v_i, v_j))$ to v_j , respectively. The variable $\widetilde{n}_1$ is a linear combination of noise terms correlated with multiple latent variables along the path. Similarly, the contribution from the confounders $l \in L$ to variable v_j , excluding v_i , is given by $\widetilde{a}_{j,l}x_l + \sum_{h \in H(P(l, v_j) \setminus v_i)} \widetilde{a}_{j,h}x_h + \widetilde{n}_2$, where $P(l, v_j) \setminus v_i$ refers to all paths from l to v_j , which do not include v_i . Thus, the data x_j can be formatted by the following structural equation:

$$\begin{aligned} x_j = & b \left(\widetilde{a}_{j,i}x_i + \sum_{h \in H(P(v_i, v_j))} \widetilde{a}_{j,h}x_h + \widetilde{n}_1 \right) \\ & + c \left(\sum_{l \in L} \widetilde{a}_{j,l}x_l + \sum_{l \in L} \sum_{h \in H(P(l, v_j) \setminus v_i)} \widetilde{a}_{j,h}x_h + \widetilde{n}_2 \right) \\ & + \sum_{p \in Pa'(v_j)} a_{j,p}x_p + n_j \end{aligned} \quad (5)$$

where $Pa'(v_j)$ denote the nodes in $Pa(v_j)$, which are not on paths $P(v_i, v_j)$ and $P(l, v_j) \setminus v_i$, n_j is the noise term for v_j , and b and c indicate the path existence. The variables b and c satisfy $b, c \in 0, 1$ and $b + c \neq 0$. Similarly, x_i is expressed as follows:

$$\begin{aligned} x_i = & c \left(\sum_{l \in L} \widetilde{a}_{i,l}x_l + \sum_{l \in L} \sum_{h \in H(P(l, v_i))} \widetilde{a}_{i,h}x_h + \widetilde{n}_3 \right) \\ & + \sum_{p \in Pa'(v_i)} a_{i,p}x_p + n_i \end{aligned} \quad (6)$$

where $Pa'(v_i)$ denote the nodes in $Pa(v_i)$ that are not on the path $P(l, v_i)$. There are three cases.

1) When $b = 1$ and $c = 1$, the relationship $v_i \rightarrow v_j$ with a common latent confounder belongs to Structure 3. The residual $r_i^{(j)}$ is calculated as the following equation:

$$\begin{aligned} r_i^{(j)} = & \sum_{l \in L} (\widetilde{a}_{i,l} - s\widetilde{a}_{j,i}\widetilde{a}_{i,l} - s\widetilde{a}_{j,l})x_l + (1 - s\widetilde{a}_{j,i})\widetilde{n}_3 \\ & + \sum_{l \in L} \sum_{h \in H(P(l, v_i))} (\widetilde{a}_{i,h} - s\widetilde{a}_{j,i}\widetilde{a}_{i,h})x_h + (1 - s\widetilde{a}_{j,i})n_i - s\widetilde{n}_1 \\ & + \sum_{p \in Pa'(v_i)} (a_{i,p} - s\widetilde{a}_{j,i}a_{i,p})x_p - s \sum_{p \in Pa'(v_j)} a_{j,p}x_p - s\widetilde{n}_2 \\ & - s \sum_{h \in H(P(v_i, v_j))} \widetilde{a}_{j,h}x_h - s \sum_{l \in L} \sum_{h \in H(P(l, v_j) \setminus v_i)} \widetilde{a}_{j,h}x_h - sn_j \end{aligned} \quad (7)$$

where $s = (\text{cov}(x_i, x_j) / \text{var}(x_j))$. If $\text{cov}(x_i, x_j) = 0$, the coefficient of $\widetilde{n}_3$ on $r_i^{(j)}$ is nonzero. Thus, $r_i^{(j)}$ is dependent of x_j according to Theorem 1. If $\text{cov}(x_i, x_j) \neq 0$, both $\widetilde{n}_1$

and $\widetilde{n}_2$ have nonzero coefficients on $r_i^{(j)}$, showing that the dependence remains.

2) When $b = 1$ and $c = 0$, the relationship $v_i \rightarrow v_j$ without any confounders belongs to Structure 1. The residual $r_i^{(j)}$ is calculated as the following equation:

$$\begin{aligned} r_i^{(j)} = & \sum_{p \in Pa'(v_i)} (a_{i,p} - s\widetilde{a}_{j,i}a_{i,p})x_p + (1 - s\widetilde{a}_{j,i})n_i - s\widetilde{n}_1 \\ & - s \sum_{h \in H(P(v_i, v_j))} \widetilde{a}_{j,h}x_h - s \sum_{p \in Pa'(v_j)} a_{j,p}x_p - sn_j. \end{aligned} \quad (8)$$

If $\text{cov}(x_i, x_j) = 0$, the coefficient of $\widetilde{n}_1$ on $r_i^{(j)}$ is nonzero. When $\text{cov}(x_i, x_j) \neq 0$, $\widetilde{n}_1$ on $r_i^{(j)}$ is nonzero. Therefore, in this situation, $r_i^{(j)}$ is still dependent of x_j due to Theorem 1.

3) When $b = 0$ and $c = 1$, v_i and v_j are only affected by confounders, belonging to Structure 2. The following is the calculation of the residual $r_i^{(j)}$:

$$\begin{aligned} r_i^{(j)} = & \sum_{l \in L} (\widetilde{a}_{i,l} - s\widetilde{a}_{j,i}\widetilde{a}_{i,l})x_l + \sum_{l \in L} \sum_{h \in H(P(l, v_i))} \widetilde{a}_{i,h}x_h + \widetilde{n}_3 \\ & + \sum_{p \in Pa'(v_i)} a_{i,p}x_p + n_i - s \sum_{p \in Pa'(v_j)} a_{j,p}x_p \\ & - s \sum_{l \in L} \sum_{h \in H(P(l, v_j) \setminus v_i)} \widetilde{a}_{j,h}x_h - s\widetilde{n}_2 - sn_j. \end{aligned} \quad (9)$$

If $\text{cov}(x_i, x_j) \neq 0$, the coefficient of $\widetilde{n}_2$ for $s = r_i^{(j)}$ is nonzero. The finite-time linear combination of non-Gaussian data x_l still meets the non-Gaussian distribution. When $\text{cov}(x_i, x_j) = 0$, the coefficient of x_l is nonzero.

In summary, $x_j \in \mathbf{x}_j$ and $r_i^{(j)} \in \mathbf{r}_i^{(j)}$ are dependent in all cases, which contradicts what is known. Hence, Proposition 1 is proven. $\square$

The HSCI method is retained for the CI test in Proposition 1. The significance parameter for the CI test in Proposition 1 is labeled as sig_2 .

2) *Proposition 2:* Proposition 1 determines whether the structure between v_i and v_j is Structure 1. Furthermore, Proposition 2 shows that the matrix $\tilde{\mathbf{B}}'$ can distinguish between Structure 2 and Structure 3. The matrix $\tilde{\mathbf{B}}'$ originates from [27] and fundamentally acts as an approximation of matrix $\mathbf{B}$. The matrix $\mathbf{B}$ is described in (3). The element of matrix $\mathbf{B}$, as discussed in [27], shows whether a causal path is present between the related variables. Thus, we can determine whether the causal path exists by using $\tilde{\mathbf{B}}'$ instead of $\mathbf{B}$. In our work, we employ the RICA algorithm to calculate $\tilde{\mathbf{B}}'$, setting the sampling parameter r to ensure result reliability [30]. The lemma in [27] gives the necessary and sufficient condition for causal path existence between variables.

Lemma 1 [27]: Given the faithfulness assumption, there is a causal path from the observed variable v_i to v_j if and only if the corresponding two-row vectors $\tilde{\mathbf{B}}'_{i,:}$ and $\tilde{\mathbf{B}}'_{j,:}$ in the matrix $\tilde{\mathbf{B}}'$ satisfy $n_{0*} > 0$ and $n_{*0} = 0$, where n_{0*} means the number of columns in $[\tilde{\mathbf{B}}'_{i,:}, \tilde{\mathbf{B}}'_{j,:}]$ whose first element is zero and the second is nonzero, and n_{*0} is the number of columns in $[\tilde{\mathbf{B}}'_{i,:}, \tilde{\mathbf{B}}'_{j,:}]$ whose first element is nonzero and the second is zero.

With Lemma 1 as a basis, we derive Proposition 2, which distinguishes Structure 2 from Structure 3 and identifies the causal direction.

Proposition 2: Given the faithfulness assumption, if two variables (v_i, v_j) at the edge G_t' are affected by the confounders, then the matrix $\tilde{\mathbf{B}}'$ can distinguish whether their structure is Structure 2 or Structure 3.

Proof: If the two variables v_i and v_j in $(v_i, v_j) \in U$ are affected by the confounders, it can be known from Property 1 that the two variables can only be Structure 2 or Structure 3. If two-row vectors $\tilde{\mathbf{B}}'_{i,:}$ and $\tilde{\mathbf{B}}'_{j,:}$ in the matrix $\tilde{\mathbf{B}}'$ satisfy $n_{0*} > 0$ and $n_{*0} = 0$, it implies the presence of a causal path between two variables according to the Lemma 1. Therefore, the structure of the two variables belongs to Structure 3, in which there is a causal path between the two variables. Conversely, it belongs to Structure 2, in which there is no causal path between the two variables. In conclusion, the matrix $\tilde{\mathbf{B}}'$ can distinguish between the two structures. $\square$

To facilitate a clear understanding of the Matrix $\tilde{\mathbf{B}}'$ and Proposition 2, we present the following examples.

Example 1: We apply the ICDPLV to the data generated according to Fig. 1(b). All the examples in the subsequent text adopt the same data. When $t = 1$, the LiNGAM-SF method constructs the skeleton structure G_1' shown in Fig. 2 for the arrived variables v_1, v_2 , and v_3 . The ICDPLV needs to perform operations on the three edges (v_1, v_2) , (v_2, v_3) , and (v_1, v_3) . It can be known from Proposition 1 that $r_1^{(3)}$ and $r_2^{(3)}$ are independent of x_3 respectively. Therefore, the directed edges $\langle v_3, v_1 \rangle$ and $\langle v_3, v_2 \rangle$ are determined. For the undirected edge (v_1, v_2) , it does not satisfy the conditions of Proposition 1, indicating that v_1 and v_2 are affected by confounders. The two vectors obtained by the RICA technique are given as follows:

$$\begin{bmatrix} 0.1036 & 1 \\ 1 & 0.1307 \end{bmatrix} \approx \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \quad (10)$$

According to Proposition 2, the relationship between v_1 and v_2 is Structure 2. Therefore, when $t = 1$, the causal graph output by ICDPLV is G_1 in Fig. 2.

Example 2: When new variables $\{v_4, v_5, v_8\}$ arrive, the structure of G_1 is updated to G_2' in Fig. 2. We execute the ICDPLV on G_2' . For the undirected edge (v_4, v_5) on G_2' , Proposition 1 detects that it is affected by the confounders and marks it as $[v_4, v_5]$. The RICA gives the vector of $\{v_4, v_5\}$ as follows:

$$\begin{bmatrix} 0.2528 & 0.8588 \\ 0.8006 & 0.7916 \end{bmatrix} \approx \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}. \quad (11)$$

The two row vectors fulfill the conditions: $n_{0*} > 0$ and $n_{*0} = 0$. Hence, the relationship between v_4 and v_5 belongs to Structure 2. The undirected edge (v_4, v_5) is updated to $\langle v_4, v_5 \rangle$.

3) ICDPLV Subalgorithm: This article proposes the ICDPLV subalgorithm based on the above two propositions. For the set of undirected edges, the ICDPLV first identifies Structure 1 by Proposition 1, and the causal direction between the variables can be identified (lines 3–11 in Algorithm 2). Next, Proposition 2 is employed to distinguish between Structure 2 and Structure 3, thereby providing the causal direction between variables (lines 12–19 in Algorithm 2). In Structure 2 and Structure 3, the latent confounders are unobservable and unknown. However, their existence can be detected, and they are saved in L_{alter} (line 9 in Algorithm 2). The details of the ICDPLV subalgorithm are demonstrated in Algorithm 2.

Algorithm 2 ICDPLV Subalgorithm

Input: The local directed graph G_t' ; The observed data X_o ; The sampling rate r ; The significant coefficients sig_2

Output: The causal structure G_t ; The candidate set of latent confounders L_{alter} ;

```

1: Initial  $T \leftarrow \emptyset$ ,  $G_t \leftarrow \emptyset$  and  $L_{\text{alter}} \leftarrow \emptyset$ ;
2: Find a set  $U$  of undirected edges from  $G_t'$ ;
3: for all  $e_i = (v_m, v_n) \in U$  do
4:   Identify the direction via Proposition1 with  $\text{sig}_2$ 
5:   if  $v_m \rightarrow v_n$  or  $v_n \rightarrow v_m$  then
6:     Update  $(v_m, v_n)$  on  $G_t'$ ;
7:   else
8:      $T \leftarrow T \cup (v_m, v_n)$ 
9:      $L_{\text{alter}} \leftarrow L_{\text{alter}} \cup [v_m, v_n]$ 
10:  end if
11: end for
12: for all  $e_i = (v_m, v_n) \in T$  do
13:   Distinguish the structures via Proposition2 with  $r$ 
14:   if  $v_m \rightarrow v_n$  or  $v_n \rightarrow v_m$  then
15:     Update  $(v_m, v_n)$  on  $G_t'$ 
16:   else
17:     Delete undirected edge  $(v_m, v_n)$  on  $G_t'$ 
18:   end if
19: end for
20:  $G_t \leftarrow G_t'$ 
21: return  $G_t, L_{\text{alter}}$ ;

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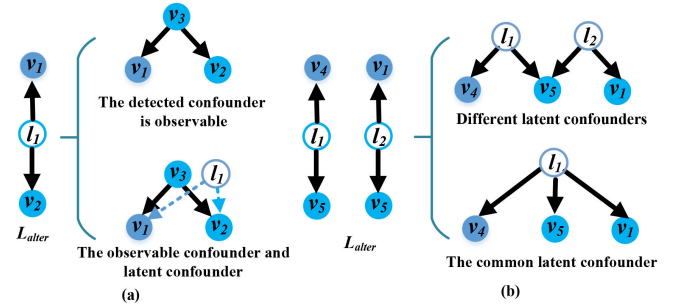


Fig. 5. Possible cases in the candidate set of latent confounders. (a) Observed confounders and latent confounders. (b) Common latent confounder influencing multiple variables.

E. DLCs Algorithm

The latent confounders L_{alter} detected by the ICDPLV subalgorithm cannot be taken as the final outcome. The reasons are given as follows. First, as shown in Fig. 5(a), the ICDPLV finds that variables v_1 and v_2 are affected by latent confounders. The ICDPLV regards the observed variable v_3 as a latent confounder because it only analyzes the relationship between two variables on an undirected edge each time. It leads to a situation where the latent confounder detected is just the observed variable v_3 in G_t , potentially along with other latent confounders such as l_1 . The situation is regular in SF scenarios, where the variable that is a latent confounder at the time t may become an observable variable at the time $t + 1$. Second, as shown in Fig. 5(b), the ICDPLV detects the presence of a confounder on the variables v_1 and v_5 , as well as on the variables v_4 and v_5 . However, v_1, v_5 , and v_6 might be affected by a common latent confounder l_1 .

TABLE II
STATISTICAL RESULTS

Real Structure	Structure with Common Latent Confounder	Fully Connected Structure
CHILD	39	0
HAILFINNDR	563	0
MILDEW	3	0
PATHFINDER	193527	4

For the case of Fig. 5(a), a solution has been given in the literature [13]. It proves theoretically that the residuals $r_i^{(j)}$ of the variable v_i can be used as a new input to eliminate the influence of observed confounder v_j . Therefore, the observed data of the variables x_i are replaced by the residuals $r_i^{(j)}$.

After removing the effect of the confounder, Proposition 1 in Section III-D will again be used to judge whether the relationship between two variables is Structure 1 unaffected by the confounders. If they belong to structure 1, the suspicious element will be eliminated from the candidate set. Otherwise, it will be retained.

Example 3: When $t = 1$, the ICDPLV outputs $L_{alter} = [v_1, v_2]$ in Fig. 2. We can observe that variables v_1 and v_2 have a common ancestor v_3 in G_1 . Therefore, the DLC needs to eliminate the influence of v_3 . The data of v_1 and v_2 are updated to $r_1^{(3)}$ and $r_2^{(3)}$, respectively. Proposition 1 makes another judgment on the new data of v_1 and v_2 . Since their new data are still related to their residuals, the variables v_1 and v_2 are still affected by the latent confounders.

The DLC subalgorithm needs to detect the common confounder, as shown in Fig. 5(b), but the task is challenging. We analyze the structure with the common confounder, as shown in Fig. 5(b). With $V_{com} = \{v_1, v_2, \dots, v_p\}$ sharing a common confounder l , the structure exhibits Property 2.

Property 2: For the structure with the common confounder, the algorithm will output $(p(p-1)/2)$ confounders of the same form as $v_i \leftarrow l_k \rightarrow v_j$, where $v_i, v_j \in \{v_1, v_2, \dots, v_p\}$ and $i \neq j$. The latent confounder is denoted as l_k , $k = 1, 2, \dots, (p(p-1)/2)$.

Proof: Any two features $v_i, v_j \in \{v_1, v_2, \dots, v_p\}$ are not directly causal, so $v_i \not\perp\!\!\!\perp v_j | l$. However, l is a latent variable, and then, v_i and v_j satisfy $\neg \exists s \subseteq V_t - \{v_i, v_j\}$ s.t. $(v_i \perp\!\!\!\perp v_j | s)$. Therefore, the method in Section III-C will create an undirected edge for v_i and v_j . Since any two of the p variables can form such an edge, $(p(p-1)/2)$ undirected edges are provided for the ICDPLV to judge. In fact, the common confounder causes ICDPLV to output $(p(p-1)/2)$ candidate latent confounders with the same form as $v_i \leftarrow l_k \rightarrow v_j$. $\square$

According to Property 2, the DLC gives an acceptable solution to detect the common latent confounders. Suppose that there are p variables in the candidate set L_{alter} , which do not have any causal relationship with each other and satisfy Property 2. In that case, the DLC will consider the p variables as having the same confounders. The p variables with $(p(p-1)/2)$ different confounders also satisfy Property 2. Later, we call this type of structure a fully connected structure for short. However, there are too few fully connected structures in the real causal structure. The statistical results are given in Table II to verify the point. Therefore, the DLC can give an acceptable result in detecting the common latent confounders.

The details of the DLC subalgorithm are shown in Algorithm 3. The main purpose of the DLC subalgorithm is to detect the latent confounders from the candidate set

given by the ICDPLV. The DLC first removes the effect of confounders from the observed variables by residuals (lines 2–17 in Algorithm 3) and then detects the common confounders based on Property 2 (lines 18–21 in Algorithm 3). Finally, the DLC outputs the detected latent confounders.

Algorithm 3 DLC Subalgorithm

Input: The causal structure G ; The candidate set of latent confounders L_{alter} ; The observed data X_o ; The latent confounders L^{t-1} output at the time $t-1$

Output: The set of latent confounders L^t ;

- 1: Initial $LC \leftarrow \emptyset$ and $L^t \leftarrow \emptyset$
- 2: $[v_1, v_2, \dots, v_n] \in L^{t-1}$ are split into $(p(p-1)/2)$ confounders
- 3: $L_{alter} \leftarrow L_{alter} \cup L^{t-1}$
- 4: **for** each (v_i, v_j) in L_{alter} **do**
- 5: Find observed confounders l_k of v_i and v_j in G
- 6: $LC(v_i) \leftarrow LC(v_i) \cup l_k$
- 7: $LC(v_j) \leftarrow LC(v_j) \cup l_k$
- 8: **end for**
- 9: $T \leftarrow$ all feature variables t in LC
- 10: **for** each t in T **do**
- 11: Compute the residual r on $LC(t)$
- 12: $X_o(:, t) \leftarrow r$;
- 13: **end for**
- 14: **for** each (v_i, v_j) in L_{alter} **do**
- 15: **if** Check v_i and v_j are not influenced by latent confounders based on **Proposition 1** **then**
- 16: $L_{alter} \leftarrow L_{alter} - (v_i, v_j)$
- 17: **end if**
- 18: **end for**
- 19: **if** The set V_{com} that satisfies **Property 2** **then**
- 20: Delete $[v_i, v_j]$ in L_{alter} , $v_i, v_j \in V_{com}$
- 21: $L^t \leftarrow L^t \cup [v_1, v_2, \dots, v_n]$
- 22: **end if**
- 23: $L^t \leftarrow L^t \cup L_{alter}$
- 24: **return** L^t ;

IV. EXPERIMENTS

A. Experimental Preparation

We conduct the experiments with a standard PC on MATLAB R2018b. It is important to note that the experiments utilized all observational data without partitioning it into training and testing sets. The approach to evaluation is to compare the learned results with the ground truth (real structure).

1) *Dataset:* 1) The synthetic data are generated based on real causal structures derived from the Bayesian Network Repository¹ and randomly generated networks. Table III summarizes these structures. To satisfy linear non-Gaussianity, data is generated following (1). The causal intensity A ranges over $[-1, 0) \cup (0, 1]$, and the noise N is uniformly distributed over $(-1, 1)$, ensuring independence and non-Gaussianity. Each structure has 1000 data samples.

2) The real data are from the Yahoo Finance database.² We obtain daily closing prices from August 30, 2013, to

¹Bayesian network repository: <https://www.bnlearn.com/bnrepository/>.

²Yahoo finance database: <https://help.yahoo.com/kb/sln2311.html>.

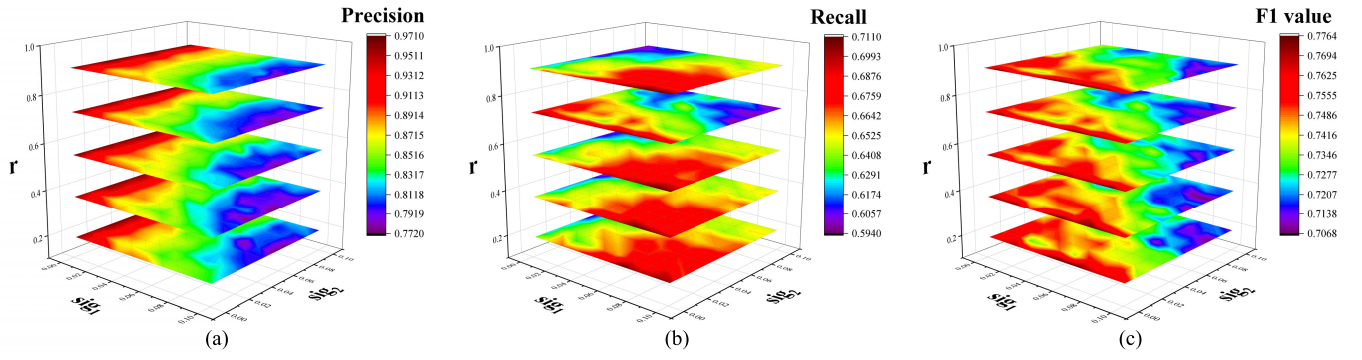


Fig. 6. Parameter sensitivity analysis. (a) Precision results for three parameters under different values. (b) Recall results for three parameters under different values. (c) F1 value results for three parameters under different values.

TABLE III

DETAILED INFORMATION OF THE CAUSAL STRUCTURE

Name	Size	Fields	Nodes	Edges	Use
CHILD	Medium	Medicine	20	25	Section IV-B
MILDEW	Medium	Agriculture	35	46	Section IV-B
HAILFINDER	Large	Meteorology	56	66	Section IV-B
PATHFINDER	Very Large	Biology	109	105	Section IV-B
SACHS	Small	Biology	11	17	Section IV-C
ALARM	Medium	Medicine	37	46	Section IV-C
BARLEY	Medium	Agriculture	48	84	Section IV-C and IV-D
INSURANCE	Medium	Insurance	27	52	Section IV-C
WATER	Medium	Ecology	32	66	Section IV-C and IV-D
HEPAR2	Large	Biology	70	123	Section IV-C and IV-D
WIN95PTS	Large	Engineering	76	112	Section IV-C and IV-D
ANDES	Large	Education	223	338	Section IV-C
DIABETES	Large	Medicine	413	602	Section IV-C
PIGS	Very Large	Culture	441	592	Section IV-C
LINK	Very Large	Biology	724	1125	Section IV-C
MUNIN	Massive	Biology	1041	1397	Section IV-C
SY1500	Massive	Randomly generated	1500	2638	Section IV-C
SY2000	Massive	Randomly generated	2000	3994	Section IV-C
SY5000	Massive	Randomly generated	5000	17486	Section IV-C

August 29, 2023, for three index funds: U.S. Dow Jones Industrial (DJI) Average, Japanese Nikkei 225 (N225), and Hong Kong Hang Seng Index (HSI). Following the data preprocessing methodology from [27], let $c(t)$ denote the closing price on day t . The return $R(t)$ is defined as: $(c(t) - c(t-1))/c(t-1)$. We utilize index returns as feature variables to investigate the causal relationships between indices. The preprocessed dataset consists of 2153 samples with three features.

2) *Comparison Methods*: The CSBS [9], the CSSU [10], and the PRCDSF [12] are current CSLSF methods for linear relationships, so they are chosen as comparison methods. To validate the advantages of partitioning the data into SFs, the one-time learning methods such as L-MARVEL [21], M³HC [23], and ParceLiNGAM [25] are also employed as comparison methods.

3) *Evaluation Metrics*: The comparison metrics of the methods contain three items: precision P , recall R , and F1 value. The precision is the percentage of correct causal relationships among learned causal relationships between two observed feature variables. The recall calculates the percentage of correct causal relationships among true causal relationships between two observed feature variables. The F1 value is calculated as $2 \times P \times R / (P + R)$.

B. Parameter Sensitivity Analysis

There are three parameters for the LiNGAM-SF method. They are the sampling rate r and two significant coefficients,

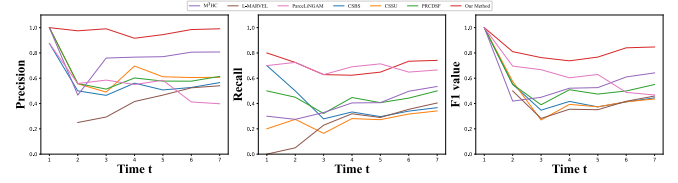


Fig. 7. Three comparison metrics on ALARM under SFs.

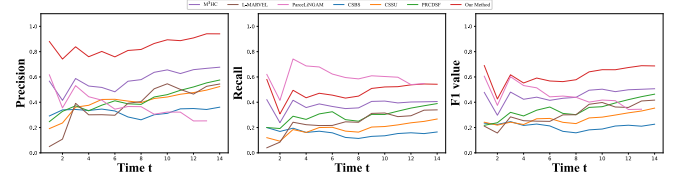


Fig. 8. Three comparison metrics on HEPAR2 under SFs.

sig_1 and sig_2 . The performance of the method in terms of the precision, recall, and F1 value is observed by letting r vary from 0 to 1 with a step size of 0.1 and letting sig_1 and sig_2 vary from 0.001 to 0.1 with a step size of 0.011. The parameter sensitivity is analyzed on the structures: CHILD, HAILFINDER, MILDEW, and PATHFINDER. The precision, recall, and F1 values in Fig. 6 are the average values obtained on the four structures.

Fig. 6(a) indicates that the smaller the sig_1 , the higher the precision. Fig. 6(b) reveals that the smaller the sig_2 , the higher the recall rate. In Fig. 6(b), when $r = 0.7$ and $\text{sig}_2 = 0.001$, the recall rate can reach above 0.66. In summary, the three parameters in the experiments are set as $r = 0.7$, $\text{sig}_1 = 0.001$, and $\text{sig}_2 = 0.001$, respectively.

C. Comparative Experiments

1) Synthetic Data:

a) *Streaming features*: To simulate the scenario of SFs, the data are partitioned into batches based on feature variables and then fed into all methods in batches. Each batch randomly includes k features, with k respectively assigned as 2, 5, 5, 10, and 10 for five sizes of causal structures: small, medium, large, very large, and massive. The results are averaged over ten experimental runs, with data regenerated for each run to meet the dataset requirements described in Section IV-A.

Figs. 7–10 present the precision, recall, and F1 values across different time points for the ALARM, HEPAR2, PIGS, and

TABLE IV
COMPARISON EXPERIMENTS ON THE COMPLETE DATA—PART A

Metrics	Methods	SACHS	ALARM	BARLEY	INSURANCE	WATER	HEPAR2	WIN95PTS
P	M ³ HC[23]	0.65 ± 0.15●	0.81 ± 0.12●	0.69 ± 0.06●	0.67 ± 0.13●	0.66 ± 0.12●	0.68 ± 0.03●	0.77 ± 0.09●
	L-MARVEL[21]	0.47 ± 0.23●	0.54 ± 0.07●	0.64 ± 0.06●	0.69 ± 0.09●	0.60 ± 0.07●	0.54 ± 0.17●	0.42 ± 0.07●
	ParceLiNGAM[25]	0.88 ± 0.06●	0.70 ± 0.16●	0.39 ± 0.07●	0.45 ± 0.14●	0.40 ± 0.09●	-	-
	CSBS[9]	0.33 ± 0.18●	0.57 ± 0.04●	0.48 ± 0.05●	0.43 ± 0.06●	0.53 ± 0.06●	0.36 ± 0.05●	0.47 ± 0.05●
	CSSU[10]	0.35 ± 0.19●	0.61 ± 0.05●	0.52 ± 0.05●	0.41 ± 0.08●	0.52 ± 0.10●	0.52 ± 0.06●	0.51 ± 0.04●
	PRCDSF[12]	0.38 ± 0.22●	0.61 ± 0.02●	0.52 ± 0.04●	0.50 ± 0.06●	0.52 ± 0.06●	0.58 ± 0.03●	0.50 ± 0.04●
	LiNGAM-SF	0.99 ± 0.03	0.99 ± 0.02	0.94 ± 0.03	0.92 ± 0.08	0.93 ± 0.04	0.94 ± 0.03	0.98 ± 0.02
R	M ³ HC[23]	0.43 ± 0.09●	0.53 ± 0.11●	0.44 ± 0.06●	0.42 ± 0.10●	0.32 ± 0.07●	0.41 ± 0.02●	0.49 ± 0.08●
	L-MARVEL[21]	0.19 ± 0.10●	0.40 ± 0.10●	0.47 ± 0.07●	0.54 ± 0.11○	0.45 ± 0.09●	0.34 ± 0.13●	0.30 ± 0.06●
	ParceLiNGAM[25]	0.87 ± 0.11○	0.66 ± 0.20●	0.63 ± 0.10○	0.65 ± 0.13○	0.65 ± 0.07○	-	-
	CSBS[9]	0.20 ± 0.12●	0.37 ± 0.03●	0.25 ± 0.04●	0.21 ± 0.04●	0.24 ± 0.03●	0.17 ± 0.02●	0.25 ± 0.03●
	CSSU[10]	0.19 ± 0.11●	0.34 ± 0.04●	0.27 ± 0.03●	0.22 ± 0.06●	0.23 ± 0.04●	0.27 ± 0.03●	0.24 ± 0.02●
	PRCDSF[12]	0.26 ± 0.17●	0.50 ± 0.05●	0.40 ± 0.04●	0.33 ± 0.03●	0.39 ± 0.06●	0.39 ± 0.03●	0.37 ± 0.02●
	LiNGAM-SF	0.70 ± 0.09	0.74 ± 0.06	0.59 ± 0.02	0.54 ± 0.07	0.54 ± 0.05	0.54 ± 0.03	0.69 ± 0.03
F1	M ³ HC[23]	0.51 ± 0.10●	0.64 ± 0.11●	0.54 ± 0.06●	0.51 ± 0.11●	0.43 ± 0.09●	0.51 ± 0.02●	0.60 ± 0.08●
	L-MARVEL[21]	0.26 ± 0.11●	0.46 ± 0.09●	0.54 ± 0.06●	0.61 ± 0.10○	0.51 ± 0.08●	0.42 ± 0.15●	0.35 ± 0.06●
	ParceLiNGAM[25]	0.87 ± 0.07○	0.66 ± 0.15●	0.48 ± 0.08●	0.53 ± 0.14●	-	-	-
	CSBS[9]	0.25 ± 0.14●	0.45 ± 0.03●	0.33 ± 0.05●	0.28 ± 0.05●	0.33 ± 0.04●	0.22 ± 0.03●	0.33 ± 0.04●
	CSSU[10]	0.25 ± 0.14●	0.44 ± 0.04●	0.36 ± 0.04●	0.28 ± 0.07●	0.32 ± 0.06●	0.35 ± 0.03●	0.33 ± 0.03●
	PRCDSF[12]	0.31 ± 0.19●	0.55 ± 0.04●	0.45 ± 0.04●	0.40 ± 0.04●	0.45 ± 0.06●	0.46 ± 0.02●	0.43 ± 0.02●
	LiNGAM-SF	0.82 ± 0.07	0.85 ± 0.04	0.73 ± 0.02	0.68 ± 0.07	0.68 ± 0.04	0.69 ± 0.02	0.81 ± 0.02
Time	M ³ HC[23]	2.8 ± 2.9	2.1 ± 1.0	4.8 ± 2.4	3.9 ± 1.8	9.7 ± 5.0	7.3 ± 1.7	8.0 ± 1.8
	L-MARVEL[21]	0.1 ± 0.0	0.2 ± 0.0	0.4 ± 0.0	0.1 ± 0.0	0.3 ± 0.0	1.4 ± 0.1	1.6 ± 0.0
	ParceLiNGAM[25]	35.9 ± 3.3	1412.5 ± 126.3	2993.4 ± 167.7	549.5 ± 25.9	935.1 ± 65.6	-	-
	CSBS[9]	0.4 ± 0.1	0.6 ± 0.1	1.4 ± 0.3	1.5 ± 0.3	0.9 ± 0.2	1.2 ± 0.4	1.1 ± 0.2
	CSSU[10]	0.2 ± 0.1	0.2 ± 0.0	0.2 ± 0.0	0.2 ± 0.0	0.2 ± 0.0	0.2 ± 0.1	0.2 ± 0.1
	PRCDSF[12]	1.1 ± 0.1	1.1 ± 0.0	1.1 ± 0.0	0.9 ± 0.1	0.9 ± 0.1	1.1 ± 0.1	1.0 ± 0.1
	LiNGAM-SF	1.5 ± 0.3	2.7 ± 0.3	4.6 ± 0.2	3.5 ± 0.4	3.5 ± 0.4	1.8 ± 0.4	3.5 ± 0.5

‘●/○/○’ denote respectively that LiNGAM-SF is significantly better/tied/worse than the compared method by the t-test with confidence level 0.05.

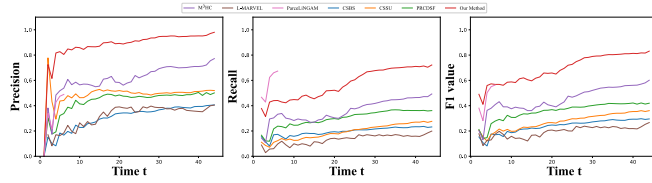


Fig. 9. Three comparison metrics on PIGS under SFs.

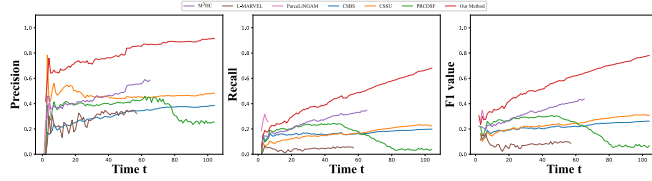


Fig. 10. Three comparison metrics on MUNIN under SFs.

MUNIN structures under SFs. Our method, represented by the red line, demonstrates consistently superior performance in terms of precision and F1 values throughout the entire temporal phase compared to all baseline methods.

The comparison algorithms M³HC, ParceLiNGAM, and L-MARVEL are representatives of causal structure methods with latent variables. The ParceLiNGAM fails to generate results within the 2-h time limit when the number of feature variables exceeds 70. Similarly, M³HC and L-MARVEL exceed this computational threshold when feature variables exceed 700. The limitation of their one-time learning prevents them from coping with the incremental features.

b) *Complete data*: Tables IV and V summarize the performance metrics for complete data across 15 structures, with “-” indicating a 2-h timeout. Compared to the best baseline, our method achieves 11% higher precision on aver-

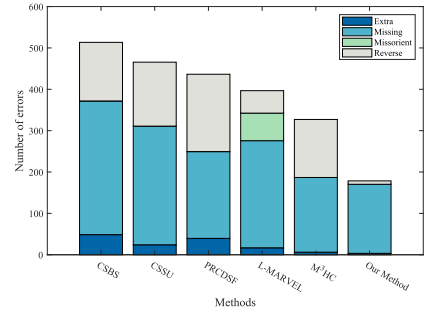


Fig. 11. Analysis on error results.

age. The performance gap widens for high-dimensional cases (>1000 variables), where our F1 values exceed state-of-the-art results by 59% on average. Compared to other methods, ParceLiNGAM achieves the highest recall on SACHS, BARLEY, INSURANCE, and WATER structures. However, it lacks the capability to learn causal structures in high-dimensional data. In higher dimensional cases, both M³HC and L-MARVEL face similar challenges. Therefore, we suggest dividing high-dimensional complete data into SFs, employing the batch-learning method to learn causal structure.

The CSBS, CSSU, PRCDSF, and M³HC methods rely on scoring to identify directions, but, as noted in Section I, this approach is unreliable. Taking the DIABETES structure as an example, these methods produce excess and reversed edges, resulting in poor outcomes (see Fig. 11). In contrast, the L-MARVEL method, which uses d-separation to identify causal directions, cannot distinguish Markov equivalence classes, leading to numerous undirected edges, as shown in Fig. 11. As shown in Fig. 12, LiNGAM-SF achieved over 90% precision in learning undirected edges, closely approximating the final outcomes. The ability to precisely learn causal

TABLE V
COMPARISON EXPERIMENTS ON THE COMPLETE DATA—PART B

Metrics	Methods	ANDES	DIABETES	PIGS	LINK	MUNIN	SY1500	SY2000	SY5000
P	M ³ HC[23]	0.80 ± 0.05●	0.66 ± 0.03●	0.77 ± 0.02●	-	-	-	-	-
	ParceLiNGAM[25]	-	-	-	-	-	-	-	-
	L-MARVEL[21]	0.52 ± 0.02●	0.62 ± 0.01●	0.40 ± 0.04●	-	-	-	-	-
	CSBS[9]	0.38 ± 0.02●	0.42 ± 0.02●	0.41 ± 0.01●	0.38 ± 0.02●	0.38 ± 0.01●	0.16 ± 0.02●	0.15 ± 0.02●	-
	CSSU[10]	0.45 ± 0.02●	0.47 ± 0.02●	0.52 ± 0.02●	0.43 ± 0.02●	0.48 ± 0.01●	0.49 ± 0.02●	0.47 ± 0.03●	0.39 ± 0.04●
	PRCDSF[12]	0.41 ± 0.01●	0.48 ± 0.01●	0.50 ± 0.01●	0.34 ± 0.03●	0.25 ± 0.03●	0.30 ± 0.02●	0.28 ± 0.03●	0.25 ± 0.01●
	LiNGAM-SF	0.97 ± 0.01	0.97 ± 0.01	0.98 ± 0.01	0.90 ± 0.02	0.91 ± 0.01	0.99 ± 0.00	0.98 ± 0.00	0.93 ± 0.01
R	M ³ HC[23]	0.52 ± 0.04●	0.47 ± 0.02●	0.49 ± 0.02●	-	-	-	-	-
	L-MARVEL[21]	0.38 ± 0.03●	0.37 ± 0.01●	0.20 ± 0.03●	-	-	-	-	-
	ParceLiNGAM[25]	-	-	-	-	-	-	-	-
	CSBS[9]	0.21 ± 0.02●	0.23 ± 0.01●	0.23 ± 0.01●	0.18 ± 0.01●	0.20 ± 0.01●	0.02 ± 0.01●	0.02 ± 0.01●	-
	CSSU[10]	0.22 ± 0.01●	0.27 ± 0.01●	0.28 ± 0.01●	0.22 ± 0.01●	0.22 ± 0.01●	0.04 ± 0.01●	0.03 ± 0.01●	0.01 ± 0.00●
	PRCDSF[12]	0.35 ± 0.01●	0.34 ± 0.01●	0.36 ± 0.01●	0.07 ± 0.02●	0.04 ± 0.01●	0.03 ± 0.01●	0.02 ± 0.01●	0.01 ± 0.00●
	LiNGAM-SF	0.67 ± 0.02	0.71 ± 0.01	0.72 ± 0.01	0.62 ± 0.01	0.68 ± 0.01	0.70 ± 0.01	0.65 ± 0.01	0.46 ± 0.00
$F1$	M ³ HC[23]	0.63 ± 0.04●	0.55 ± 0.03●	0.60 ± 0.02●	-	-	-	-	-
	L-MARVEL[21]	0.43 ± 0.03●	0.46 ± 0.01●	0.27 ± 0.04●	-	-	-	-	-
	ParceLiNGAM[25]	-	-	-	-	-	-	-	-
	CSBS[9]	0.27 ± 0.02●	0.29 ± 0.02●	0.30 ± 0.01●	0.25 ± 0.01●	0.26 ± 0.01●	0.03 ± 0.01●	0.03 ± 0.01●	-
	CSSU[10]	0.30 ± 0.01●	0.34 ± 0.01●	0.36 ± 0.01●	0.29 ± 0.01●	0.31 ± 0.01●	0.07 ± 0.02●	0.05 ± 0.01●	0.02 ± 0.00●
	PRCDSF[12]	0.38 ± 0.01●	0.40 ± 0.01●	0.42 ± 0.01●	0.11 ± 0.02●	0.07 ± 0.02●	0.05 ± 0.02●	0.05 ± 0.01●	0.03 ± 0.00●
	LiNGAM-SF	0.79 ± 0.02	0.82 ± 0.01	0.83 ± 0.01	0.73 ± 0.01	0.78 ± 0.01	0.82 ± 0.01	0.79 ± 0.00	0.62 ± 0.01
Time	M ³ HC[23]	70.1 ± 9.8	517.6 ± 7.5	1275.4 ± 19.5	-	-	-	-	-
	L-MARVEL[21]	65.1 ± 1.8	579.5 ± 6.4	733.2 ± 7.8	-	-	-	-	-
	ParceLiNGAM[25]	-	-	-	-	-	-	-	-
	CSBS[9]	7.0 ± 0.5	16.3 ± 3.7	17.2 ± 1.1	52.3 ± 4.6	106.6 ± 21.7	111.1 ± 1.8	210.3 ± 7.5	-
	CSSU[10]	1.1 ± 0.1	1.7 ± 0.1	2.3 ± 0.1	6.3 ± 0.4	12.8 ± 0.7	5.6 ± 0.3	14.1 ± 0.6	15.7 ± 1.5
	PRCDSF[12]	2.5 ± 0.1	6.0 ± 0.1	5.9 ± 0.1	8.4 ± 0.1	11.6 ± 0.2	27.0 ± 0.7	67.3 ± 1.1	158.4 ± 6.7
	LiNGAM-SF	6.8 ± 0.3	12.7 ± 0.8	11.3 ± 0.5	27.8 ± 0.7	42.0 ± 3.8	57.6 ± 1.5	95.8 ± 2.8	6098.8 ± 22.07

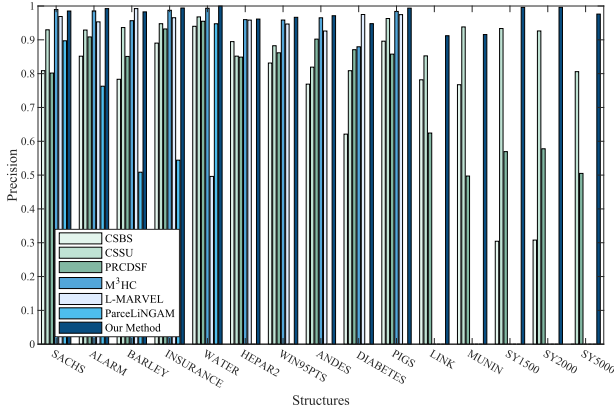


Fig. 12. Results on the precision of the learned undirected edges.

structures through causal identifiability is a key factor in the method's success.

The time efficiency of each method is recorded in Tables IV and V. Although our method shows little time advantage with fewer features, it becomes more efficient as feature numbers grow, reusing prior regression results rather than recalculating. This allows it to outperform other comparison methods. With hundreds of features, it ranks third in time usage but yields the most precise causal structure, making the time tradeoff worthwhile.

2) *Real Data*: For real data, the return values of three indices funds HSI, N225, and DJI are observed respectively at the time $t = 1, 2, 3$. The output of each method at each time is shown in Fig. 13. The existing CSLSF method establishes a causal relationship for indices HIS and N225 when $t = 2$, but its correctness is yet to be proven. In contrast, the proposed method finds that there is a latent confounder for the two indices. The conclusion is explainable because the index DJI affects indices N225 and HIS. The conclusion is a consensus

in the financial industry. Moreover, when $t = 3$, only the proposed method gives the same results as in the literature [27] and [31].

D. Latent Confounders Detection for SFs

1) *Synthetic Data*: For synthetic data, this section employs four cases to illustrate that the LiNGAM-SF method can detect latent confounders.

Case 1: For the structure WIN95PTS, the variable *Problem6* is assumed to be observed at the time $t = 1$. Next, two feature variables *GrbldPS* and *AvlblVrtlMmry* are observed at the time $t = 2$. The real causal structure is shown in Fig. 14(a).

When the variables *GrbldPS* and *AvlblVrtlMmry* are observable, the ICDPLV outputs the structure *AvlblVrtlMmry* → *GrbldPS* → *Problem6* for three variables. Meanwhile, it provides the candidate set of latent confounders [*Problem6*, *GrbldPS*]. Since *Problem6* and *GrbldPS* do not share a common confounder in the structure and do not satisfy Property 2, [*Problem6*, *GrbldPS*] is the output of the DLC. In all figures of this section, the latent confounders are consistently denoted as L .

Case 2: For the structure BARLEY, it is assumed that the two variables *exptgens* and *forfrugt* are observed at the time $t = 1$. Then, *potnmin* is observed at the time $t = 2$. The real structure is shown in Fig. 15(a).

At the time $t = 1$, the LiNGAM-SF method outputs the causal structure *forfrugt* → *potnmin*. After observing the feature variable *potnmin*, the ICDPLV updates the causal structure to *potnmin* ← *forfrugt* → *potnmin* and outputs the candidate set [*potnmin*, *exptgens*]. Unlike Case 1, *potnmin* and *exptgens* have an observed confounder *forfrugt*. Therefore, the DLC needs to remove the effect of the confounder *forfrugt* and then judge whether *potnmin* and *exptgens* are still under the influence of the latent confounder through Proposition 1. Eventually, the algorithm detects that *potnmin* and *exptgens* are still affected

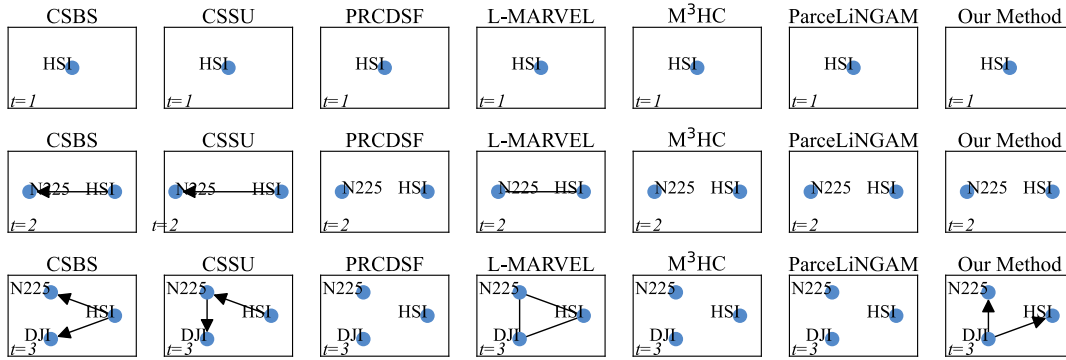
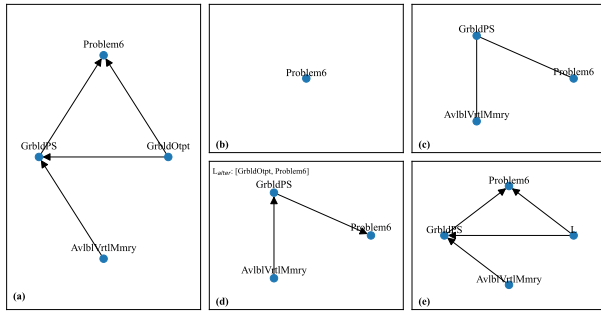
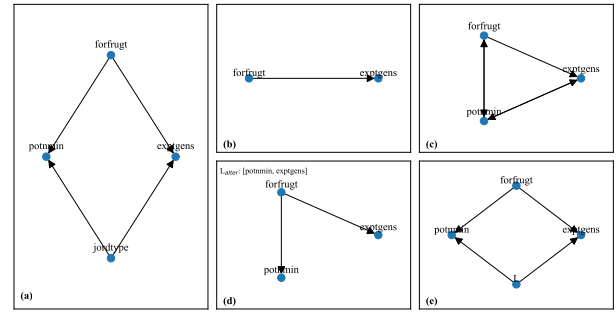


Fig. 13. Results on the real data.

Fig. 14. Latent confounder detection is performed on Case 1. (a) Real Structural. (b) $t = 1$. (c) $t = 2$ after step 1. (d) $t = 2$ after step 2. (e) $t = 2$ after step 3.Fig. 15. Latent confounder detection is performed on Case 2. (a) Real Structural. (b) $t = 1$. (c) $t = 2$ after step 1. (d) $t = 2$ after step 2. (e) $t = 2$ after step 3.

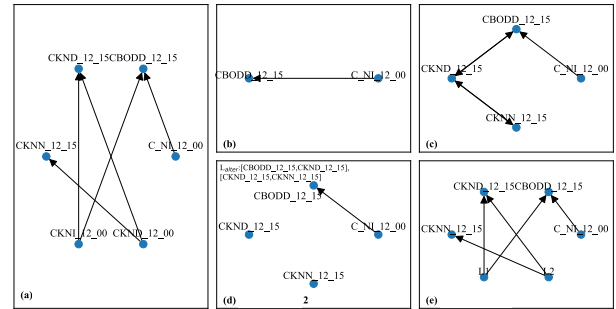
by the latent confounder L except for the influence of *forfrugt*, as shown in Fig. 15.

Case 3: For the structure WATER, it is assumed that two variables $C_NI_12_00$ and $CBODD_12_15$ are observed when $t = 1$. Subsequently, $CKND_12_15$ and $CKNN_12_15$ are observed when $t = 2$. The real structure is shown in Fig. 16(a).

The LiNGAM-SF method outputs the causal structure $C_NI_12_00 \rightarrow CBODD_12_15$ at the time $t = 1$. After observing the feature variables $CKND_12_15$ and $CKNN_12_15$, the LiNGAM-SF first learns two undirected edges: $(CKNN_12_15, CKND_12_15)$ and $(CKND_12_15, CBODD_12_15)$. Then, the ICDPLV subalgorithm outputs the candidate sets $[CKNN_12_15, CKND_12_15]$ and $[CKND_12_15, CBODD_12_15]$. It states that the latent confounders cause the two undirected edges to be generated. Thus, observed feature variables $CKNN_12_15$ and $CKND_12_15$ are isolated nodes, while the causal structure $C_NI_12_00 \rightarrow CBODD_12_15$ is retained anyhow. Since the three feature variables $CKNN_12_15$, $CKND_12_15$, and $CBODD_12_15$ in the candidate sets do not satisfy Property 2, the DLC subalgorithm finally concludes that they do not have the common latent confounder. Two latent confounders L_1 and L_2 are detected, as illustrated in Fig. 16(e).

Case 4: For the structure HEPAR2, it is assumed that two feature variables *amylase* and *fat* are observed at the time $t = 1$. Next, *upper_pain* is observed when $t = 2$. The real structure is shown in Fig. 17(a).

When $t = 1$, the LiNGAM-SF successfully detects a latent confounder $[amylase, fat]$ between *amylase* and *fat*. With the

Fig. 16. Latent confounder detection is performed on Case 3. (a) Real Structural. (b) $t = 1$. (c) $t = 2$ after step 1. (d) $t = 2$ after step 2. (e) $t = 2$ after step 3.

variable *upper_pain* observed when $t = 2$, the undirected edges $(upper_pain, amylase)$ and $(upper_pain, fat)$ are formed. Since the ICDPLV finds that they belong to Structure 2, the candidate set contains $[upper_pain, amylase]$, $[upper_pain, fat]$, and $[amylase, fat]$ output at the time $t = 1$. They are fed into the DLC for further identification. The variables in the candidate set do not have an observed confounder and conform to property 2. Thus, *amylase*, *fat*, and *upper_pain* have the common latent confounder L , as shown in Fig. 17(e).

2) Real Data: For real financial data, this section assumes that three indices funds HIS, N225, and DJI enter the observation in turn. The LiNGAM-SF method gives the latent confounder detection results for each time, as shown in Fig. 18.

When $t = 1$, only one variable HIS is observed, and the proposed method cannot give more information. As the N225

TABLE VI
ABLATION EXPERIMENT WITH TWO PROPOSITIONS

Metrics	Methods	SACHS	ALARM	BARLEY	INSURANCE	WATER	HEPAR2	WIN95PTS	ANDES	DIABETES	PIGS	LINK	MUNIN	SY1500	SY2000	SY5000
Precision	LiNGAM-SF_Del1	0.97	0.96	0.92	0.85	0.85	0.86	0.97	0.93	0.96	0.98	0.86	0.94	0.99	0.98	0.92
	LiNGAM-SF_Del2	0.98	0.99	0.98	0.93	0.93	0.97	0.97	0.98	0.98	0.98	0.95	0.92	0.99	0.99	0.94
	LiNGAM-SF	0.99	0.99	0.94	0.92	0.93	0.94	0.98	0.97	0.97	0.98	0.90	0.91	0.99	0.98	0.93
Recall	LiNGAM-SF_Del1	0.32	0.15	0.22	0.21	0.14	0.15	0.22	0.23	0.24	0.20	0.20	0.22	0.23	0.22	0.19
	LiNGAM-SF_Del2	0.60	0.72	0.48	0.44	0.39	0.49	0.59	0.58	0.63	0.69	0.54	0.65	0.62	0.55	0.41
	LiNGAM-SF	0.70	0.74	0.59	0.54	0.54	0.54	0.69	0.67	0.71	0.72	0.62	0.68	0.70	0.65	0.46
F1 Value	LiNGAM-SF_Del1	0.47	0.27	0.36	0.33	0.24	0.26	0.35	0.37	0.38	0.34	0.33	0.36	0.37	0.36	0.31
	LiNGAM-SF_Del2	0.74	0.83	0.64	0.60	0.55	0.65	0.74	0.73	0.77	0.81	0.69	0.76	0.76	0.71	0.57
	LiNGAM-SF	0.82	0.85	0.73	0.68	0.68	0.69	0.81	0.79	0.82	0.83	0.73	0.78	0.82	0.79	0.62

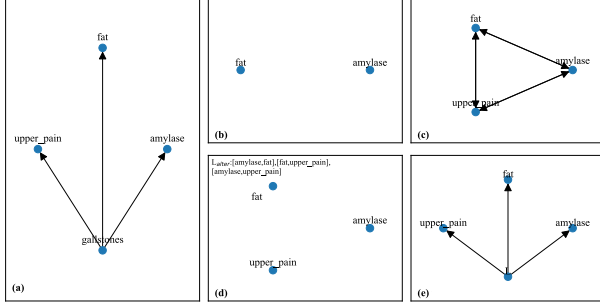


Fig. 17. Latent confounder detection is performed on Case 4. (a) Real Structural. (b) $t = 1$. (c) $t = 2$ after step 1. (d) $t = 2$ after step 2. (e) $t = 2$ after step 3.

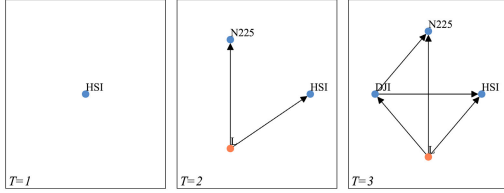


Fig. 18. Latent confounder detection is performed on real data.

index is observed, the LiNGAM-SF detects the existence of a latent confounder. The DJI index confirms the plausibility of the latent confounder when $t = 3$. The proposed method finds that the latent confounder is still present when $t = 3$. In addition, the latent confounder L is common to all three indices. In reality, it could be an event or a policy that can cause a common change in DJI, HIS, and N225 indexes. The situation is consistent with the perception of financial markets.

E. Ablation Experiment

The ICDPLV subalgorithm applies Propositions 1 and 2 to differentiate between the three structures and identify causal directions. To assess the impact of these propositions, we conduct ablation experiments, removing components based on Proposition 1 (LiNGAM-SF_Del1) and Proposition 2 (LiNGAM-SF_Del2). Table VI presents the results when all feature variables are observed.

The results of the LiNGAM-SF_Del1 method reveal the importance of Proposition 1. The LiNGAM-SF_Del1 that does not use Proposition 1 to learn the structures declined at precision, recall, and F1 values.

LiNGAM-SF_Del2 shows that excluding Proposition 2 significantly reduces recall while barely improving precision. Proposition 2 takes a further step on the result of Proposition 1 to distinguish between structures. The recall rate will either

increase or remain constant, regardless of the LiNGAM-SF_Del2's correctness.

In summary, Proposition 1 is concerned with the precision, while Proposition 2 focuses on the recall. Therefore, Propositions 1 and 2 of the ICDPLV are indispensable.

V. CONCLUSION

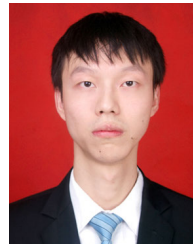
In this article, we investigate linear non-Gaussian data and propose the LiNGAM-SF method for SFs to learn a precise causal structure and detect latent confounders. The proposed method designs the ICDPLV subalgorithm to precisely identify the direction of the observed variables and output a candidate set of latent confounders. In addition, the proposed method provides the DLC subalgorithm based on Property 2 to further detect the latent confounders.

Although the method demonstrates significant advantages under SFs, future research should focus on boosting efficiency by refining CI testing and implementing distributed deployment. In addition, domain transfer can be used to enhance generalization, with causal structures from one source guiding the learning of SFs in different domains.

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